

13



a

i $\frac{1}{16}$

ii 1

iii 16

- b None. For all values of x , it will have a positive value.
- c It gets larger, approaching infinity.
- d It gets smaller, approaching zero.

14

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$\frac{1}{243}$	$\frac{1}{81}$	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27	81	243	59 049

- b As x increases, the function increases at a faster and faster rate.
- c All real x
- d $y > 0$

15

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	32	16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{1024}$

- b As x increases, the function decreases at a slower and slower rate.
- c 1
- d All real x
- e $y > 0$

16

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$-\frac{1}{32}$	$-\frac{1}{16}$	$-\frac{1}{8}$	$-\frac{1}{4}$	$-\frac{1}{2}$	-1	-2	-4	-8	-16	-32	-1024

- b No. For $a > 0$ the expression a^x is greater than zero for all real x . So, $-(a^x)$ will be less than or equal to zero.
- c Decreasing
- d As x increases, the function decreases at a faster and faster rate.
- e All real x .
- f $y < 0$

17



- a Positive
- b 0
- c $y = 0$ is a horizontal asymptote of the curve.

18

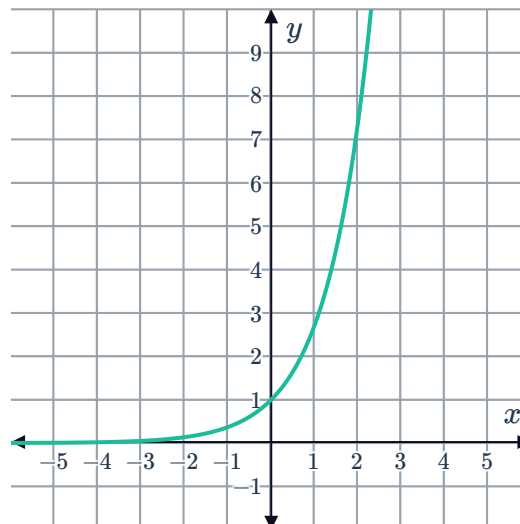
- a $(0, 1)$
- b No
- c All real x
- d $y > 0$
- e $y = 128$
- f $x = 8$

19

a

x	-3	-2	-1	0	1	2	3
$f(x)$	0.05	0.14	0.37	1.00	2.72	7.39	20.09

b



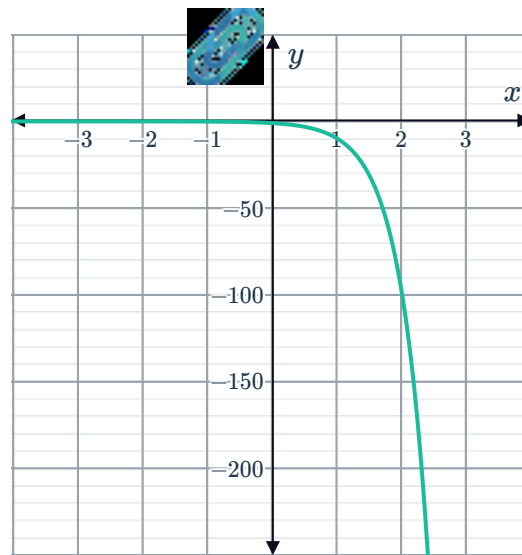
20

a

x	-2	-1	0	1	2	3
y	$-\frac{1}{100}$	$-\frac{1}{10}$	-1	-10	-100	-1000

- b Yes. For any value of x , 10^x is always positive. Therefore, -10^x is always negative.

c



21

a No. For $a > 0$ the expression a^x is greater than zero for all real x .

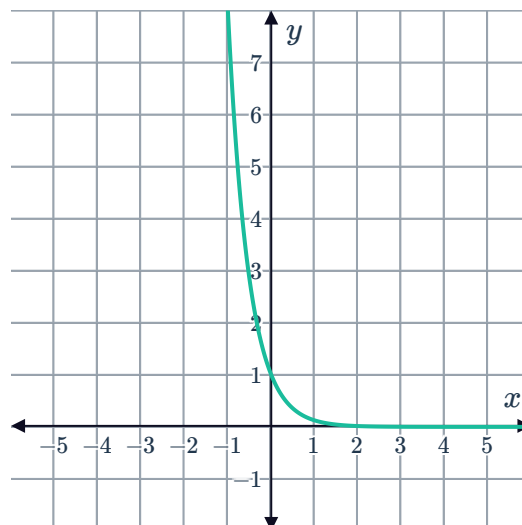
b 0

c ∞

d 1

e 0

f



22

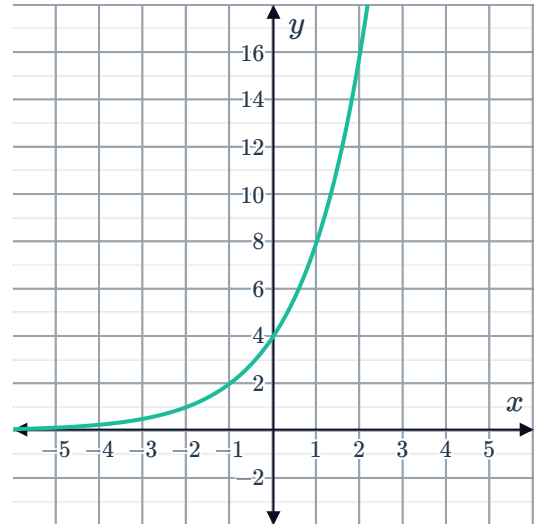
a $(0, 4)$

c ∞



b No

d



23

a 2

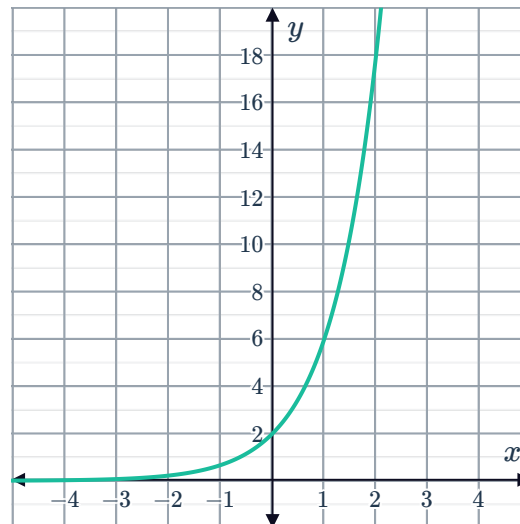
b No. For $a > 0$ the expression a^x is greater than zero for all real x . So, $2 \times (a^x)$ will be greater than or equal to zero.

c All real x .

d $y > 0$

e ∞

f



24

a Graph R

b For $x < 0$, the graph of $y = 6^x$ is below the graph of $y = 4^x$. This is because negative values of x result in fractional function values, and for any negative value of x , 6^x will result in a smaller value than 4^x .

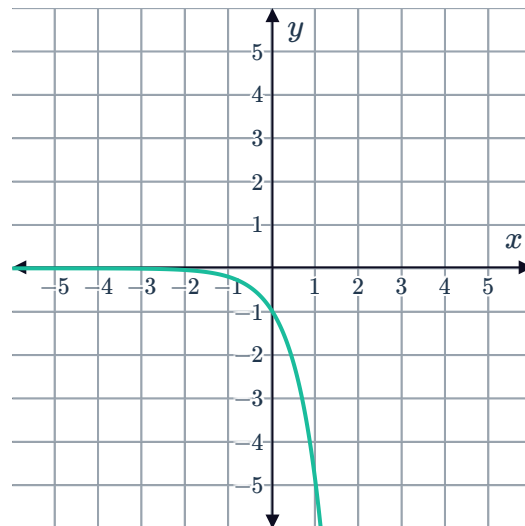
- 25 a** For $x > 0$, the graph of $y = e^x$ will lie above the graph of $y = (2)^x$ and below the graph of $y = (3)^x$.
- b** For $x < 0$, the graph of $y = e^x$ will lie above the graph of $y = (3)^x$ and below the graph of $y = (2)^x$.

Transformations of exponential functions

26

- a** A reflection over the x -axis

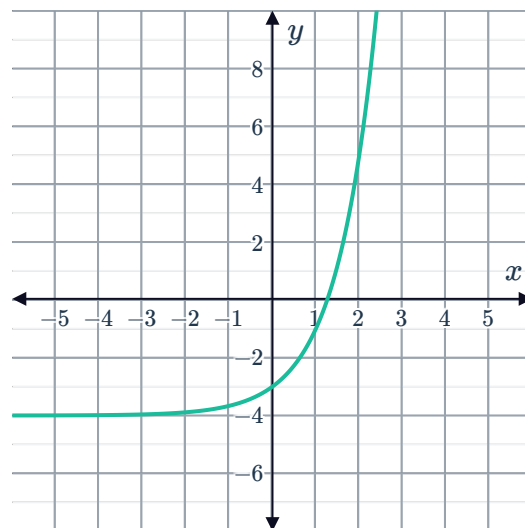
b



27

- a** Vertical translation

b

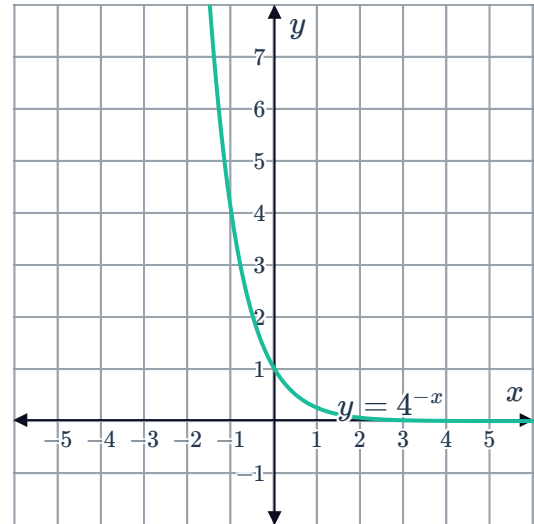


28

a Reflection over the y -axis.



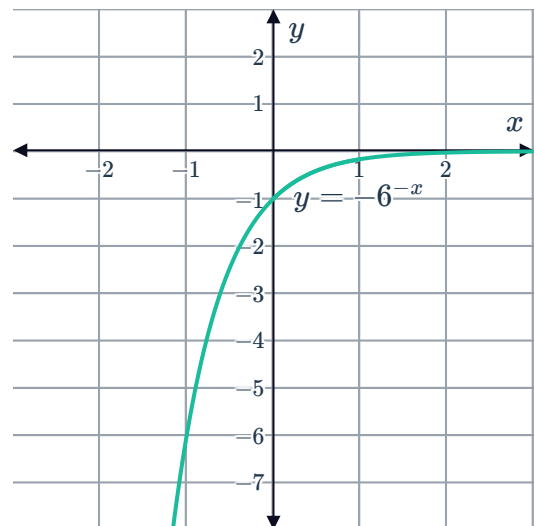
b



29

a Reflection over the x -axis and y -axis.

b



30

a $y = 0$

b $y = 2$

c 1

d All real x .

e $y < 2$

31

a 1

b -1

c $y = 0$

d $y = -2$

32



a

i 4 units to the left

ii 2 units

iii 1

b

i 1 units to the left

ii 1 units down

iii $\frac{1}{3}$

c

i $\frac{1}{5}$ units to the right

ii 2 units

iii 5

33

a $y = 2^x - 7$

b $y = 11^{x+10}$

34

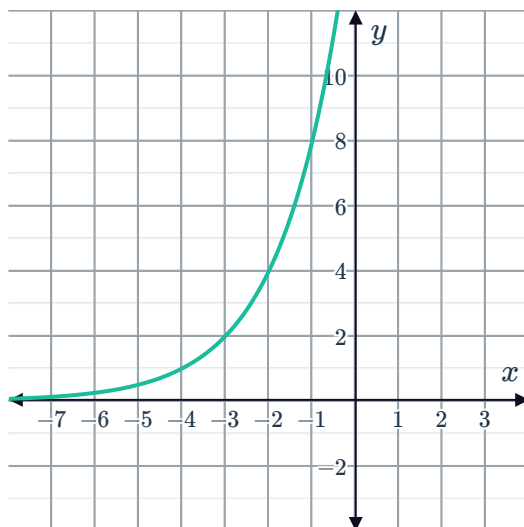
a $y = 7^{x-3} - 5$

b $y = -7^{x+3} + 5$

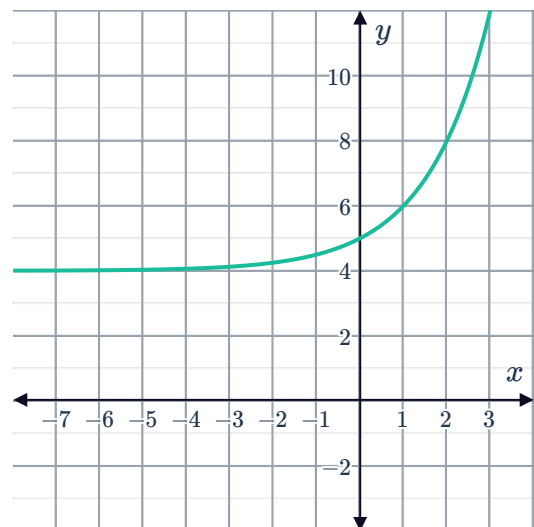
c $y = -7^{x+3} - 5$

35

a



b



36

a

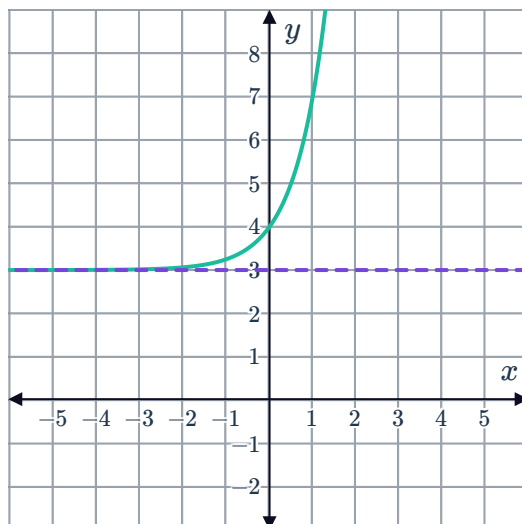
i 4

iii $y > 3$



ii All real x

iv



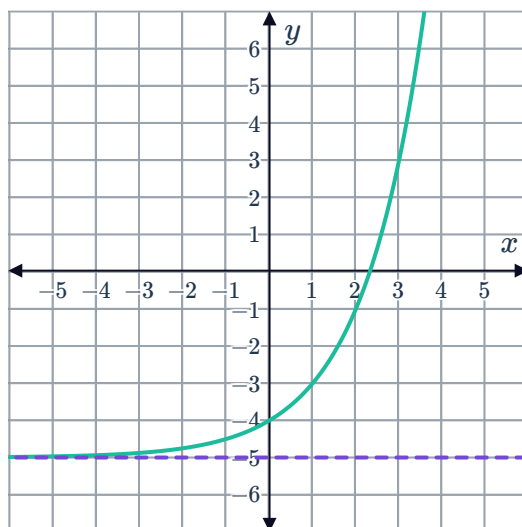
b

i -4

iii $y > -5$

ii All real x

iv



c

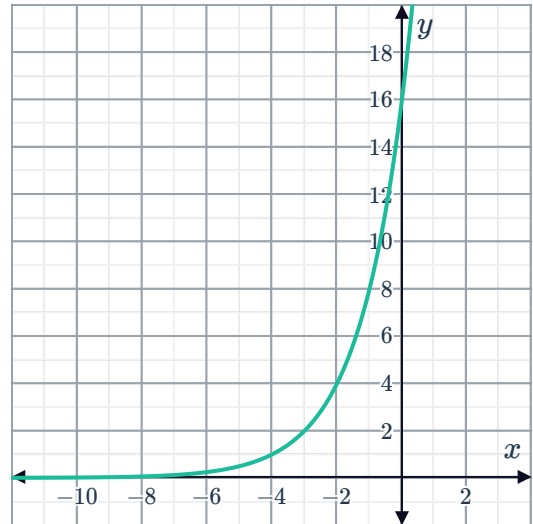
i 16

iii $y > 0$



ii All real x

iv



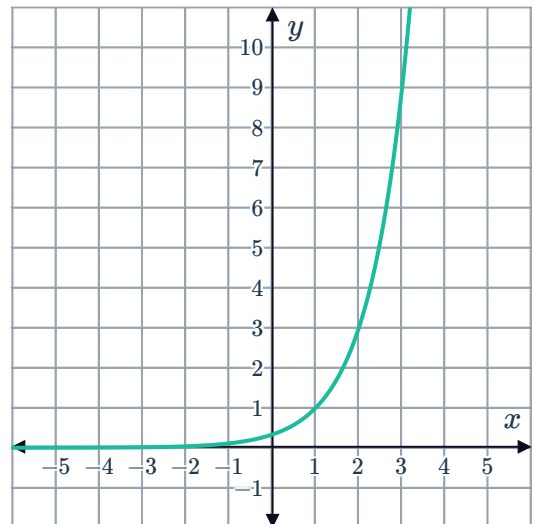
d

i $\frac{1}{3}$

iii $y > 0$

ii All real x

iv



e

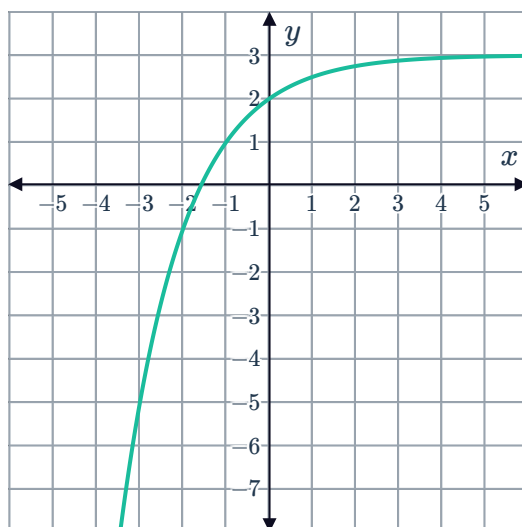
i 2

iii $y < 3$



ii All real x

iv



37

a $y = 0$

b $y = 2$

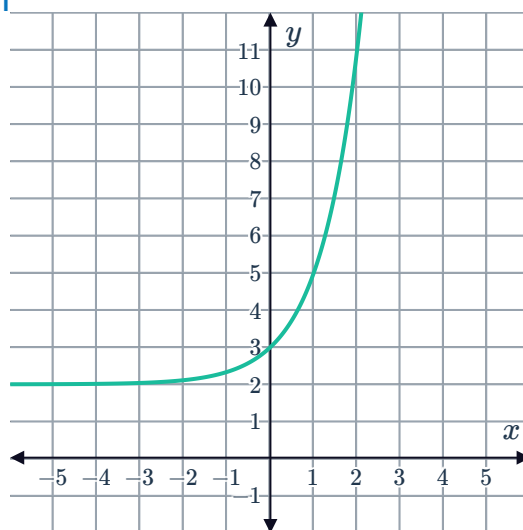
c 1

38

a i $(0, 3)$

ii $y = 2$

iii



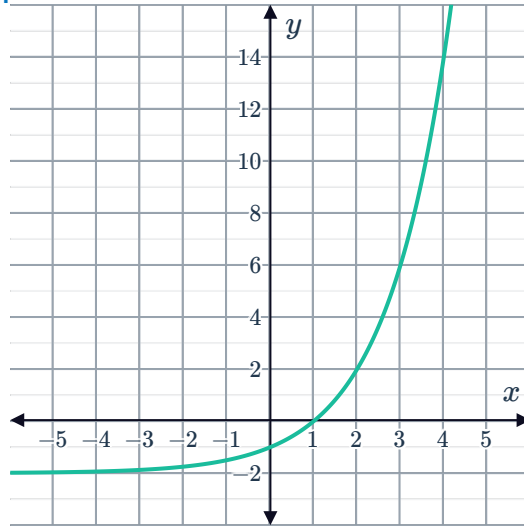
b

i $(0, -1)$

ii $y = -2$



iii

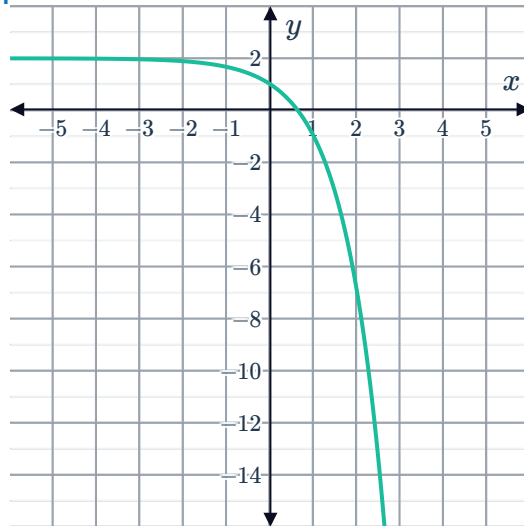


c

i $(0, 1)$

ii $y = 2$

iii

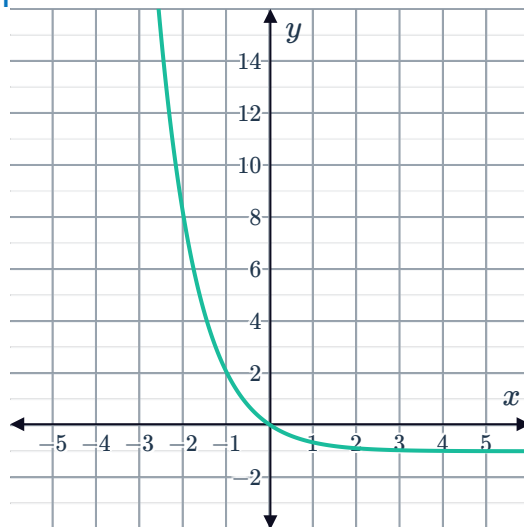


d

i $(0, 0)$

ii $y = -1$

iii



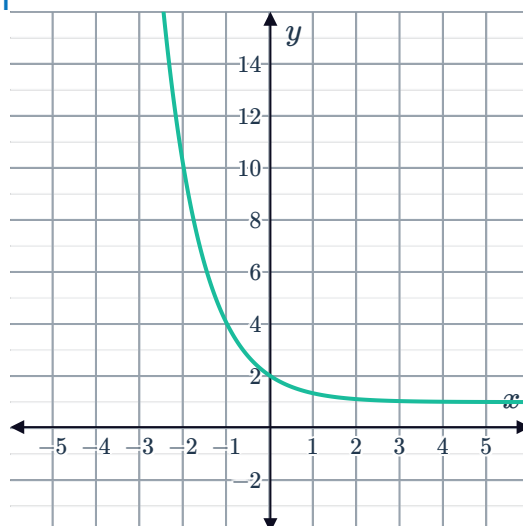
e

i $(0, 2)$

ii $y = 1$



iii



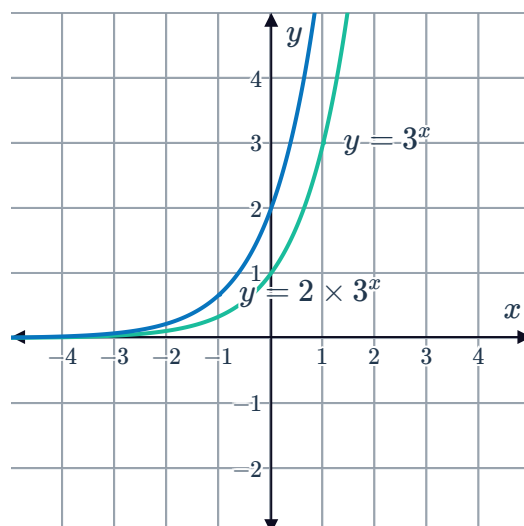
39

a

Point on original graph	Point on new graph
$\left(-1, \frac{1}{3}\right)$	$\left(-1, \frac{2}{3}\right)$
$(0, 1)$	$(0, 2)$
$(1, 3)$	$(1, 6)$
$(2, 9)$	$(2, 18)$

b $y = 2 \times 3^x$

c



d For all values of x , $y = 2 \times 3^x$ lies above $y = 3^x$.

40

a

- i No ii No iii No

c

- i Yes ii Yes iii Yes



b

- i No ii Yes iii Yes

d

- i No ii Yes iii Yes

Exponential equations

41

a $y = 2^x$

b $y = 2^{-x}$

c $y = 9^x$

d $a = 5$

42

a

x	1	2	3	4
y	2	4	8	16

b 2

c $y = 2^x$

43

a 10

b $y = 10^x$

c 10 000 000 000

44

a $a = 5$

b $b = 4$

c $y = \frac{5}{16}$